

KORONA and KoronaScript: a programmable toolbox for acoustics data processing

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Editor: [✉](#)

Submitted: 25 March 2026

Published: unpublished

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Summary

KoronaScript is a Python library that provides a programmatic interface to KORONA. KORONA is a library that contains a wide collection of algorithms and processing modules for acoustic data processing. KORONA is integrated with the Large Scale Survey System (LSSS) (Korneliussen et al., 2006), a Java based application for scientific analysis of data from acoustic equipment, including echo sounders, sonars, and hydrophones.

Statement of need

Acoustic instruments have become essential tools for marine science, and a variety of echo sounders and sonars are routinely used to explore the world below the ocean surface. A variety of acoustic instruments serve specific purposes. For instance, multi-beam echo sounders map the sea floor in high detail, split beam and multi frequency echo sounders measure the abundance of biomass in the water column, side-scan and synthetic aperture sonars can resolve objects in minute detail, and ACDP measures deep sea currents.

In parallel with the development of new and more advanced instruments, there has risen a need for algorithms and tools to effectively process the often large amounts of data produced. Software analytics are important for tasks like noise detection and removal, data compression, bottom detection, species identification and other acoustic target classification, and integrating with related data. Several popular software packages exist, often combining a variety of analytics with an easy to use graphical interface, including LSSS (Korneliussen et al., 2006), HERMES and Movies3d (Trenkel et al., 2009), and EchoView (Echoview Software Pty Ltd, 2025).

While interactive scrutiny by expert users remains an important method to interpret acoustic data, it is often convenient or even necessary to automate the processing. As data collection volumes increase, labor costs and expert availability become major obstacles to effective data use. Having a programmatic interface that can be scripted is quickly becoming a necessity.

Furthermore, many important algorithms and processing modules have many tunable parameters, and a scriptable API allows these parameters to be optimized by an automated (or semi-automated) process using grid search or other schemes. Users can perform sensitivity analysis to evaluate the importance of each parameter, and can compare the results from different configurations to find the process that best fits a particular challenge.

Finally, by embedding the analysis in a program, it can be shared, copied, and versioned (e.g. in GitHub), which supports reproducibility and verifiability, and is particularly useful for scientific work. The process can also be intermixed with other processing operations, like AI-based classification models (e.g., Brautaset et al., 2020), or functionality offered by other analysis toolkits like PyEchoView (Wall et al., 2018) or EchoPype (Lee et al., 2024).

KoronaScript is a Python application programming interface (API) that interfaces with KORONA, the processing components of the popular acoustic analysis package, LSSS. This gives the user full programmatic access to all the processing modules and algorithms offered by KORONA through a convenient Python interface.

State of the field

Traditionally, interpreting marine acoustic data for scientific or resource management purposes has been performed by manual scrutiny by experts using interactive tools. This approach has been a cornerstone for important long-running survey series performed by research institutes, examples include the Institute of Marine Research in Norway using LSSS, IFREMER in France using Movies3d, and NOAA in the US using EchoView.

To support post-processing of data and scientific information extraction, some programmatic packages exist, including EchoViewR (Harrison et al., 2015), which provides an R interface to the EchoView system, and Matecho (Perrot et al., 2018), a tool for data analysis implemented in Matlab, and linked to Movies3d (Trenkel et al., 2009).

With the advent of uncrewed platforms for data collection (Handegard et al., 2024), the need to build automated workflows has increased. Increased data complexity and volumes, e.g. from broadband echo sounders, exacerbate this (Guidi et al., 2020; Malde et al., 2020). To automate data processing, there exist packages like PyEchoLab (Wall et al., 2018) and EchoPype (Lee et al., 2024) which offer algorithms and tools with a programmatic interface.

Simultaneously, the advent of deep learning in marine science (Beyan & Browman, 2020; Malde et al., 2020; Rubbens et al., 2023) has increased the importance of integrating traditional or existing methods with new, data driven approaches. Although support for machine learning can be found in many programming languages (including Matlab and R), Python is overwhelmingly the most popular choice. We foresee that the field will increasingly move towards automation that integrates traditional and new methods using Python as the primary glue.

Software design

LSSS and KORONA

The Large Scale Survey System (LSSS) (Korneliussen et al., 2006) is an efficient open source tool for analyzing data from acoustic trawl surveys. The system is designed to support the standard workflow for acoustic trawl surveys where the data is manually curated during the survey, using an interactive interface shown in Figure 1. The system contains support for reading a wide range of data formats, a graphical interface to manually assign acoustic backscatter to acoustic categories as well as an echo integration step for estimating the nautical scattering coefficient per acoustic category for further use in survey estimation programs like StoX (Johnsen et al., 2019). The software is the *de facto* standard for all acoustic trawl surveys conducted by the Institute of Marine Research.

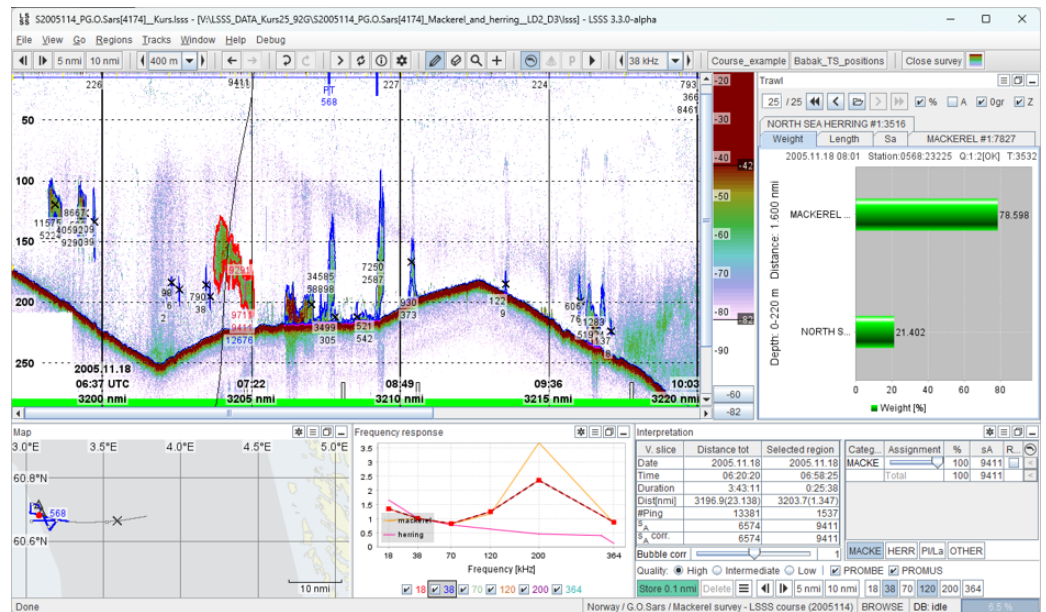


Figure 1: Interpretation of echogram by means of LSSS. The red region in the echogram is connected to the three lower windows and show that the backscatter is interpreted to originate from mackerel. This is supported by the frequency-response (middle lower window) and catch of pelagic trawl PT568 (upper right window).

LSSS integrates with a preprocessing library called KORONA. KORONA is used to process the data before using the data in LSSS. The library is comprised of a number of individual modules (see Table 1 for the full list), offering access to a wide range of functionality and algorithms, including noise removal, broadband pulse compression, tracking, automated target classification (see Figure 2), format conversions, etc. In addition to the integration with the LSSS platform, KORONA also provides its own graphical user interface for configuring and orchestrating the various processing modules. The standard output from KORONA uses a data format based on extending the raw data format used by Kongsberg, but the library also offer data conversion, and supports importing from and exporting to NetCDF.

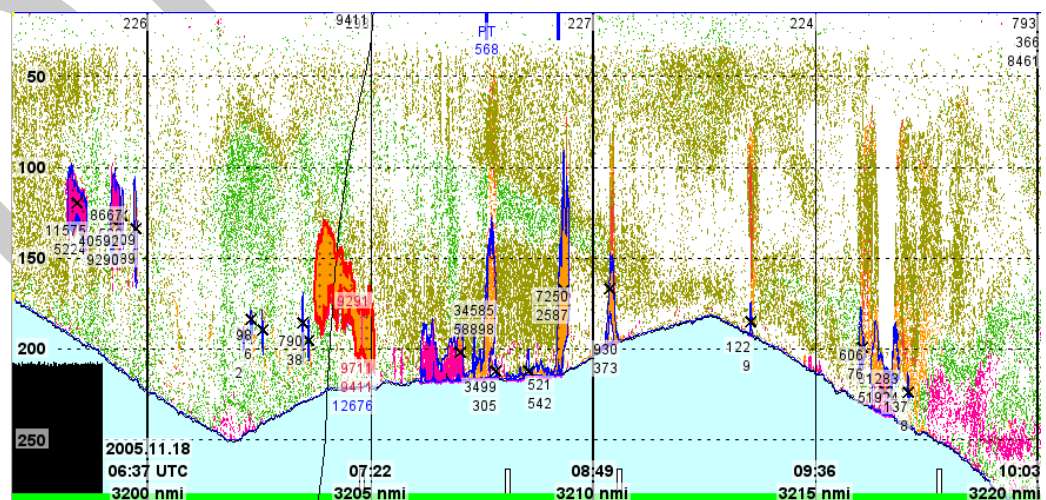


Figure 2: Result of automatic species identification by means of a selection of KORONA modules.

Table 1: The set of modules offered in the current version of KORONA, and thus available for use in Python programs with KoronaScript.

List of modules		
AngleDeletion	ErodeLowValues	PulseCompressionFilter
BroadbandNotchFilter	Erode	RemoveBottom
BroadbandSplitter	Expression	Rescale
BubblSpikeFilter	FillMissingData	SchoolCategorization
Categorization	Filter3X3	SchoolDetection
CdsViewer	FiskViewDisplay	Smoother
ChannelDataRemoval	GroupEnd	SpikeFilter
ChannelRemoval	HorizontalOffsetCorrect	SpotNoise
Combination	Isolation	TemporaryComputationsBegin
Comment	Median	TemporaryComputationsEnd
ComplexToReal	NetcdfWriter	ThresholdAllChannels
DataReduction	NoiseAcceptance	Threshold
DepthDependentResampling	NoiseMedianQuantificati	TimeInterval
Depth	NoiseQuantification	Towfish
Dilate	NoiseRemover	TrackFilter
Downsampling	NoiseVisualization	Tracking
ES60Correction	PingCollapsing	TsDetection
EchoLineCompression	PingThinning	VerticalOffsetCorrection
EdgeDetection	PlanktonInversion	Writer
EmptyPingRemoval	Plugin	

The KoronaScript programmatic interface

KoronaScript is implemented as a set of Python classes that each encapsulates a KORONA processing module. These classes leverage two core features offered by KORONA. First, KORONA can generate a description of its modules and their parameters in the form of a JSON file. KoronaScript parses this file and generates the corresponding Python classes. This ensures that the KoronaScript will automatically incorporate new features provided by new releases of KORONA.

Second, the KORONA user interface writes configured pipelines to XML files, which are then processed by the corresponding processing modules. KoronaScript leverages this interface by having the classes generate the same XML output as the KORONA GUI would produce. A KoronaScript class lets the user organize class instances in a sequence, and calling the KORONA processing infrastructure on the generated XML specification.

Availability and installation

KoronaScript is available from the PYPI archive, and can be installed using `pip install koronascript`, or via other package management tools for Python (e.g. Anaconda or uv). The source code is available from the GitHub repository at <https://github.com/CRIMAC-WP4-Machine-learning/CRIMAC-KoronaScript>.

KoronaScript will automatically download KORONA, LSSS, and the required Java runtime from the official web site and install them in a default location, so no explicit action is required by the user. If a separate installation to a user defined location is preferred, the latest version of the software can be downloaded from <https://marec.no/downloads.htm> or from the GitHub repository at <https://github.com/marec-open-source/lsss>. The environment variable LSSS must then be configured to point to the installation of these packages for KoronaScript to utilize them.

Research impact statement

KORONA and LSSS have grown in popularity over time, and before it was released as Open Source it was licensed for use by 22 institutes and other organizations¹. At the Institute of Marine Research, LSSS is used to process acoustic data from more than sixty surveys per year, and KORONA functionality is instrumental for acoustic target classification, noise filtering, data regridding, and format conversion.

KoronaScript makes the analyses easily available programmatically, thus enabling systematic exploration of methods and parameters and facilitating deployment in new settings including on UAVs and in the cloud. At the Institute of Marine Research, KoronaScript is rapidly becoming an indispensable tool for automating processing pipelines, and restructuring and ingestion of historical data for developing new machine learning methods.

AI usage disclosure

A large language model was used to review drafts and offer suggestions for text improvement. All suggestions were manually reviewed. AI has not been used for direct generation of code or text.

Acknowledgments

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